Explainable Loan Prediction Using Financial Behaviour Consistency Engine (FBCE)

S. Prem Kumar
Department of AIML
Vels Institute of Science, Technology and Advanced Studies (VISTAS)
Chennai, India

A. Guhan
Department of AIML
Vels Institute of Science, Technology and Advanced Studies (VISTAS)
Chennai, India

Dr. N. Meenakshi
Guide
Department of AIML
Vels Institute of Science, Technology and Advanced Studies (VISTAS)
Chennai, India

***Abstract*—Machine learning models can screen loan applications quickly, but their decisions are hard to justify, and they rarely test whether the financial details in an application agree with one another. This paper presents the Financial Behaviour Consistency Engine (FBCE), a decision-support prototype that places a benchmarked loan-approval classifier next to a rule-assisted consistency engine and an explanation layer based on SHapley Additive exPlanations (SHAP). Logistic regression, random forest, XGBoost and LightGBM were compared on the Loan Prediction dataset of 614 applications with stratified five-fold cross-validation; logistic regression gave the highest accuracy (80.30%) and F1-score (87.14%), with an area under the receiver operating characteristic curve of 0.751, on par with the ensembles, and was deployed. Independently of the model, FBCE scores income consistency, debt consistency, employment stability, credit consistency and repayment capacity, averages them into a 0–100 score and raises contradiction flags. A Streamlit dashboard presents the prediction, the consistency score and the signed SHAP contribution of each feature. In a case study, the model approved an applicant with a probability of only 50.09% while FBCE rated the profile at 90.89, and SHAP exposed a negative contribution from the gender attribute. Showing the three outputs together helps reviewers question borderline or biased predictions instead of accepting a bare label.**

Keywords*—*explainable artificial intelligence, loan approval prediction, SHAP, credit risk assessment, financial consistency scoring, logistic regression, decision support systems

I. Introduction

Lenders receive a steady flow of loan applications, and every decision affects both the applicant and the lender's exposure to risk. Manual appraisal of income, employment, credit history and repayment ability is thorough but slow, and two officers can reach different conclusions on the same file. Lenders therefore need tools that give consistent first-level assessments while leaving the final judgement with people.

Supervised learning is the usual response. Classifiers such as logistic regression, random forest, XGBoost and LightGBM learn from past decisions and have been applied to loan approval with encouraging results [1]–[3]. A predicted label, however, explains nothing on its own. An applicant who is refused wants to know why, and a lender has to justify automated decisions to auditors and regulators [4]–[6]. Post-hoc attribution methods such as SHapley Additive exPlanations (SHAP) [7] and Local Interpretable Model-agnostic Explanations (LIME) [8] address part of this problem by showing how each input moved a particular prediction.

A second gap receives less attention. Income, requested amount, loan term and credit history reach a classifier as separate columns, and nothing in the training objective asks whether they make sense together. A large request against a modest household income, or a clean credit record next to a heavy repayment burden, is exactly the mismatch a loan officer looks for, yet a classifier can overlook it when such combinations are rare in the training data. Explaining the model does not close this gap, because an explanation only describes what the model has learned.

This paper describes the Financial Behaviour Consistency Engine (FBCE), an academic prototype that examines an application in three ways: a benchmarked classifier predicts the outcome and its probability, a rule-assisted engine scores the internal consistency of the financial profile on five components and raises contradiction flags, and a SHAP layer explains the classifier's output feature by feature. A Streamlit dashboard shows the three views side by side, so a reviewer can see what the model predicts, why it predicts it and whether the application itself holds together.

The contributions of this work are a benchmark of four classifiers under one preprocessing pipeline and stratified five-fold cross-validation that also reports specificity; a five-component consistency engine that scores the raw application independently of the classifier and returns a bounded score, a band and contradiction flags; an explanation layer that turns SHAP attributions into ranked positive and negative drivers; and a working dashboard with a case study in which the combined outputs expose a borderline decision and a fairness concern. Sections II–VII cover related work, method, implementation, results, discussion and limitations, and conclusions.

II. Related Work

Work relevant to FBCE covers Shapley-value explanations of credit models, classifier benchmarks for loan approval and hybrid systems that pair a model with rule-based checks. Table I summarises the studies reviewed here (LR: logistic regression; RF: random forest).

Shapley values have become a common way of explaining credit models. Bussmann et al. [4] explained a gradient-

TABLE I. SUMMARY OF RELATED WORK

Ref.	Data	Classifier(s)	Explanation	Consistency or rule layer
[1]	Application, previous-application and bureau tables	LR, RF, XGBoost (with SMOTE)	–	–
[2]	Loan approval data, German credit check	RF, XGBoost, stacking, voting	LIME, SHAP	Gender fairness audit only
[3]	Kaggle loan prediction data	RF and XGBoost soft voting	SHAP, LIME	–
[4]	P2P lending, about 15,000 SMEs	XGBoost, LR	Shapley values, similarity networks	–
[5]	Norwegian bank consumer loans	LightGBM, LR scorecard	SHAP	–
[6]	Loan application data	Calibrated XGBoost	SHAP	Heuristic penalty merged into model score
[9]	German credit, 1,000 records	LR, RF, XGBoost, MLP	SHAP, LIME	–
This work	Loan Prediction, 614 applications	LR, RF, XGBoost, LightGBM	SHAP (LIME check)	Separate five-component score with contradiction flags

boosted model for about 15,000 small and medium enterprises (SMEs) seeking peer-to-peer (P2P) credit and grouped the borrowers through similarity networks built over the explanations. de Lange et al. [5] showed that LightGBM with SHAP outperformed a Norwegian bank's logistic-regression scorecard on the areas under the receiver operating characteristic (ROC) and precision-recall curves, and traced default mainly to credit-balance volatility, remaining credit and relationship length. Gawande et al. [3] tuned a soft-voting ensemble of random forest and XGBoost on a Kaggle loan prediction dataset and interpreted it with both SHAP and LIME.

On small or noisy loan data, complex models do not always win. Putra and Rumini [1] found that logistic regression gave the highest area under the ROC curve (AUC), 0.741, on merged application and bureau data balanced with the synthetic minority oversampling technique (SMOTE). Oyeyemi et al. [9] found it the most accurate model (77.3%) on the German credit data, ahead of random forest, XGBoost and a multilayer perceptron (MLP). Ghahremani et al. [2] reached 85.96% accuracy with a tuned random forest and reported disparate-impact ratios above 0.80 for all models in a gender audit.

Shah et al. [6] come closest to this work. Their system adjusts the confidence of a calibrated XGBoost model with rule-based penalties derived from financial heuristics such as the debt-to-income ratio, checks declared income against bank statements with optical character recognition (OCR) and uses SHAP for interpretation. Because the rules change the model score, a reviewer cannot tell whether a low value came from the model or from a rule. As Table I shows, explanation methods are now routine, but they explain only the model's own prediction; none of the surveyed systems reports a separate measure of internal consistency, and the one hybrid design [6] folds its rules into the model score. FBCE keeps the two apart and presents the consistency score, with its components and flags, next to the model probability.

III. PROPOSED METHODOLOGY

Fig. 1 outlines the prototype. The raw application follows two paths. It is preprocessed, scored by the selected classifier and explained with SHAP, and, in parallel, the consistency engine evaluates the same raw values with transparent rules. The decision layer collects the three results for the dashboard.

TABLE II. ATTRIBUTES OF THE LOAN PREDICTION DATASET

Attribute	Type and values	Missing	FBCE
ApplicantIncome	Numeric, monthly	0	✓
CoapplicantIncome	Numeric, monthly	0	✓
LoanAmount	Numeric, in thousands	22	✓
Loan_Amount_Term	Numeric, months	14	✓
Credit_History	1 = meets guideline, 0 = not	50	✓
Self_Employed	Yes, No	32	✓
Gender	Male, Female	13	–
Married	Yes, No	3	–
Dependents	0, 1, 2, 3+	15	–
Education	Graduate, Not graduate	0	–
Property_Area	Urban, Semiurban, Rural	0	–
Loan_Status	Target: Y (422), N (192)	0	–

A. Dataset and Preprocessing

The experiments use the public Loan Prediction dataset [10], which contains 614 labelled applications with eleven input attributes and a binary outcome (Table II). Of these, 422 applications (68.7%) were approved and 192 (31.3%) were rejected, and seven attributes have missing values.

Exploratory analysis showed how strongly credit history shapes the outcome. Among records with a recorded value, 79.6% of applicants who met the credit guideline were approved, compared with 7.9% of those who did not. Self-employment made almost no difference: 68.3% of self-employed and 68.6% of salaried applicants were approved.

Numeric attributes are imputed with the median and standardised as

$$z_j = \frac{x_j - \mu_j}{\sigma_j}, \tag{1}$$

where μ_j and σ_j are the mean and standard deviation of feature j in the training fold. Categorical attributes are imputed with their most frequent category and one-hot encoded. Both steps sit inside a single scikit-learn column transformer that is refitted in every training fold, so no statistics leak from the test fold. The transformer prefixes numeric outputs with `num__` and categorical indicators with `cat__`; these names also appear in the dashboard.

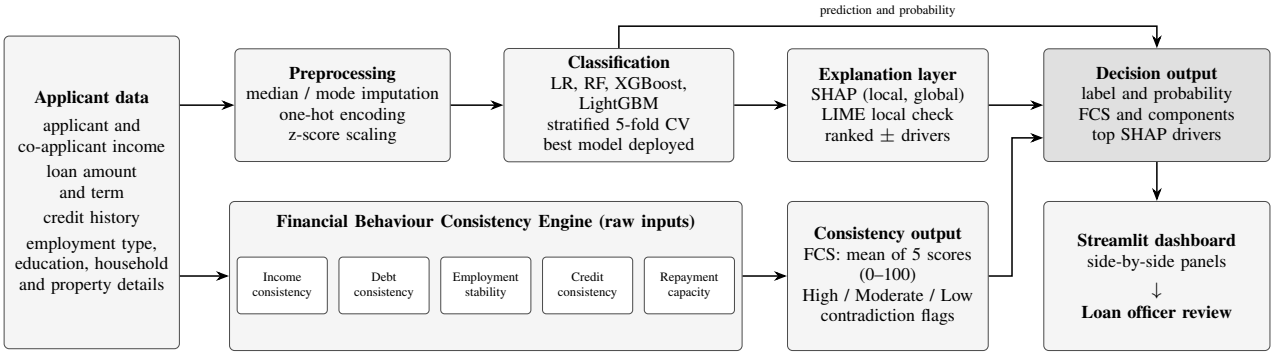


Fig. 1. Architecture of the FBCE prototype. The consistency engine reads the raw application in parallel with the machine learning path, and its output (FCS: financial consistency score) is reported next to the model prediction rather than inside it.

B. Classification Pipeline

Four classifiers are compared: logistic regression (LR), random forest (RF), XGBoost and LightGBM. For LR the approval probability is

$$p(y = 1 \mid \mathbf{z}) = \sigma(\mathbf{w}^\top \mathbf{z} + b) = \frac{1}{1 + e^{-(\mathbf{w}^\top \mathbf{z} + b)}}, \qquad (2)$$

where $\mathbf{z}$ is the transformed input, $\mathbf{w}$ the weight vector and b the bias. LR uses L2 regularisation with $C = 1$. RF uses 300 trees with at least two samples per leaf. XGBoost uses 300 boosting rounds of depth-3 trees, a learning rate of 0.05 and row and column subsampling of 0.9, and LightGBM uses 300 rounds with 15 leaves and a learning rate of 0.05. No hyperparameter search was carried out, so the comparison reflects sensible defaults rather than tuned optima.

Each model is evaluated with stratified five-fold cross-validation (shuffled, seed 42), with approval as the positive class. Besides accuracy, precision, recall, the F1-score and the AUC, we report specificity, $TN/(TN+FP)$, which measures how many of the historically rejected applications the model would also reject. The model with the highest mean F1-score is deployed, and the AUC is checked to make sure that the choice does not cost ranking quality.

C. Financial Behaviour Consistency Engine

FBCE works on the raw application rather than on the transformed features, so each of its outputs can be traced to the values the applicant entered. Let I_A and I_C be the applicant's and co-applicant's monthly incomes, $I_T = I_A + I_C$ the household income, L the requested loan amount, T the term in months, $C \in \{0, 1\}$ the credit-history flag and $E \in \{0, 1\}$ the self-employment flag. Two ratios summarise the request:

$$\rho = \frac{L}{I_T}, \qquad m = \frac{\hat{R}}{I_T}, \qquad (3)$$

where ρ is the loan-to-income ratio in the dataset's native units and $\hat{R}$ is the estimated monthly instalment for L over T months, so that m is the share of monthly income absorbed by repayment. With $\mathrm{clip}(u) = \min\{100, \max\{0, u\}\}$, the five component scores are

$$S_{\text{inc}} = \mathrm{clip}(100 - \alpha_1 \rho), \qquad (4)$$

$$S_{\text{debt}} = \mathrm{clip}(100 - \alpha_2 \rho), \qquad \alpha_2 > \alpha_1, \qquad (5)$$

$$S_{\text{emp}} = \begin{cases} s_E, & E = 1, \\ s_S, & E = 0, \end{cases} \qquad (6)$$

$$S_{\text{cred}} = \begin{cases} c_1, & C = 1, \\ c_0, & C = 0, \end{cases} \qquad (7)$$

$$S_{\text{rep}} = \mathrm{clip}\big(100\,[1 - \psi(m)]\big). \qquad (8)$$

Income consistency (4) penalises requests that are large relative to household income. As the dataset records no existing liabilities, debt consistency (5) uses the requested loan as the debt proxy with a steeper penalty. Tenure is also missing, so employment stability (6) relies on the employment type: self-employed applicants receive a lower level ($s_E = 65$ in the prototype) than salaried ones ($s_S > s_E$), reflecting the assumed volatility of business income. Credit consistency (7) maps the credit-history flag to a fixed level ($c_1 = 90$ for a positive history), and repayment capacity (8) applies a penalty $\psi(m) \in [0, 1]$ that stays near zero while the instalment is a small share of income and rises as the share approaches a project-defined tolerance. The sensitivities α_1 and α_2, the levels s_S and c_0 and the shape of ψ were set by the project team and have not yet been fitted to repayment outcomes.

The Financial Consistency Score (FCS) is the unweighted mean of the components,

$$\mathrm{FCS} = \tfrac{1}{5}\big(S_{\text{inc}} + S_{\text{debt}} + S_{\text{emp}} + S_{\text{cred}} + S_{\text{rep}}\big), \qquad (9)$$

and it is mapped to a High, Moderate or Low consistency band with two cut-offs $\theta_L < \theta_H$. Alongside the score, a small rule base $\mathcal{R}$ checks for contradictions between attributes, such as a positive credit history combined with an unusually heavy loan burden, and every rule that fires adds a readable flag to the output. The FCS is a project-defined consistency indicator, not a credit score, and the dashboard says so explicitly.

Algorithm 1 FBCE Decision Support for One Application

Input: raw application $\mathbf{x}$, fitted transformer $\mathcal{T}$, classifier h, background set $\mathcal{B}$, rule base $\mathcal{R}$
Output: label $\hat{y}$, prob. p, FCS, band, flags Φ, contributors

$\mathbf{z} \leftarrow \mathcal{T}(\mathbf{x})$ {impute, encode, scale}
$p \leftarrow h(\mathbf{z})$; $\hat{y} \leftarrow$ APPROVED if $p \geq 0.5$, else REJECTED
compute I_T, ρ and m from $\mathbf{x}$ using (3)
compute $S_{\text{inc}}, \ldots, S_{\text{rep}}$ using (4)–(8)
FCS $\leftarrow$ mean of the five scores (9); assign the band
$\Phi \leftarrow \{\, r \in \mathcal{R} : r(\mathbf{x}) \text{ is true} \,\}$
$\boldsymbol{\phi} \leftarrow \text{SHAP}(h, \mathbf{z}, \mathcal{B})$; keep the ten largest $|\phi_j|$
split the kept values into positive and negative contributors
return $\hat{y}$, p, FCS, band, $\{S_k\}$, Φ, contributors

D. Explainability Layer

SHAP [7] writes a prediction $f(\mathbf{x})$ as an additive sum of feature contributions,

$$f(\mathbf{x}) = \phi_0 + \sum_{j=1}^{M} \phi_j, \tag{10}$$

where ϕ_0 is the expected model output over a background set and ϕ_j is the Shapley value of feature j,

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!\,(M - |S| - 1)!}{M!} \big[f_{S \cup \{j\}} - f_S \big]. \tag{11}$$

Here F is the set of M features and f_S is the model output when only the features in S are known. For the deployed LR model, SHAP is computed on the log-odds output. If the features are treated as independent, the values reduce to

$$\phi_j = w_j \,(z_j - \mathbb{E}[z_j]), \tag{12}$$

so each explanation is exact and needs only one pass over the weights. Averaging $|\phi_j|$ over the training data gives a global ranking, while for a single application the dashboard lists the ten largest $|\phi_j|$ as positive (towards approval) and negative (towards rejection) contributors. LIME [8], which fits a sparse linear surrogate around the application, serves as an independent local check.

E. Decision Output and Dashboard

The decision layer does not merge the three signals into one number. It returns (Algorithm 1) the label and approval probability, the FCS with its band, components and flags, the household income used by the rules and the ranked SHAP contributors, with a note asking the reviewer to check the FCS and SHAP factors before relying on the prediction. The dashboard collects the eleven inputs through a form, shows the loaded model and a note that the FCS is not a banking credit score in its sidebar, and presents these outputs in separate panels.

TABLE III. STRATIFIED FIVE-FOLD CROSS-VALIDATION RESULTS (FOLD MEANS)

Model	Acc. (%)	Prec. (%)	Rec. (%)	Spec. (%)	F_1 (%)	AUC
LR	80.30	79.01	97.16	43.28	87.14	0.751
RF	79.15	78.84	95.26	43.81	86.26	0.749
XGBoost	78.18	79.53	91.94	47.98	85.25	0.752
LightGBM	75.40	79.00	87.43	48.97	82.97	0.740

IV. IMPLEMENTATION

The prototype is written in Python 3 with pandas and NumPy, scikit-learn, the XGBoost, LightGBM and SHAP libraries, Matplotlib and Seaborn, and Streamlit for the interface. It was developed in VS Code and Jupyter Notebook on a laptop with an Intel Core i5 or AMD Ryzen 5 class processor and 8 GB of RAM. The fitted pipeline is saved after training and loaded when the app starts, so each submission is scored, checked and explained on the fly. The app runs locally in a browser, with no connection to any banking system.

V. RESULTS

A. Classifier Comparison

Table III lists the fold means. LR achieved the highest accuracy (80.30%), recall (97.16%) and F1-score (87.14%), and its AUC (0.751) practically equalled those of XGBoost (0.752) and RF (0.749). Fold standard deviations of 1.1 to 5.3 percentage points exceed most gaps between models, so the four classifiers are close on data of this size, in line with the logistic-regression results in [1] and [9]. LR was deployed for its F1-score and because (12) makes its SHAP values exact and cheap to compute.

Specificity is less comfortable. No model rejected even half of the historically rejected applications (from 43.28% for LR to 48.97% for LightGBM), and the boosted models gained a little specificity only by losing recall. Across the test folds, LR approved 109 of the 192 rejected applications, and 89.0% of those had a positive credit flag. The classifiers thus lean heavily on credit history, which matches the approval gap noted in Section III-A. This is where an independent consistency check helps, because FBCE judges affordability and coherence rather than past approval patterns.

B. Case Study

Table IV gives the inputs and outputs for the case-study applicant, while Fig. 2 shows the ten largest SHAP contributions.

The classifier approved the application with a probability of 50.09%, only 0.09 percentage points above the threshold. FBCE was far more certain: three components exceeded 99.7, credit consistency was 90 and employment stability 65, giving an FCS of 90.89 (High) with no contradiction flags for a household income of 6,500. The one weak component comes from the self-employment rule. In the SHAP view,

TABLE IV. CASE-STUDY INPUTS AND PROTOTYPE OUTPUTS

Input	Value	Output	Value
Applicant income	5000	Prediction	Approved
Co-applicant income	1500	Approval probability	50.09%
Loan amount (thousands)	150	Income consistency	99.77
Loan term (months)	360	Debt consistency	99.71
Credit history	Good (1)	Employment stability	65.00
Self-employed	Yes	Credit consistency	90.00
Education	Graduate	Repayment capacity	99.97
Gender, married	Male, Yes	FCS	90.89
Dependents	0	Consistency band	High
Property area	Urban	Contradiction flags	None

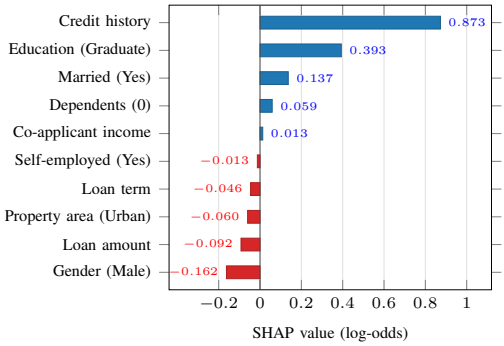


Fig. 2. Signed SHAP contributions of the ten most influential features for the case-study applicant, as reported by the dashboard (the `num__` and `cat__` prefixes are omitted). Positive values push the prediction towards approval.

the positive credit history added 0.873 to the log-odds and graduate education 0.393, while male gender (-0.162) and the loan amount (-0.092) were the largest negative terms.

VI. DISCUSSION AND LIMITATIONS

The case study leads to three observations. First, the signals differ sharply in confidence: the label alone suggests a routine approval and the probability alone a coin toss, while the FCS adds that nothing in the application is incoherent and the SHAP list shows what the model weighed. Together they support approval after review. The same layout flags the reverse case, a confident approval with a low FCS or an active flag, which the specificity results suggest is common.

Second, the explanation exposed a gender effect. Male gender was the third-largest contributor and lowered the log-odds by 0.162, so the model has absorbed a historical pattern that should play no part in credit approval and that, without SHAP, would have stayed hidden behind an approval. Gender, and possibly marital status, should therefore be dropped from the features and disparate impact audited as in [2].

Third, the two layers disagree about self-employment. FBCE penalises it by design, the classifier gives it almost no weight (-0.013) and the data show nearly identical approval rates for both groups, so the rule encodes a lending assumption that should be calibrated against repayment outcomes. Likewise, the three ratio-based components all exceeded 99.7 here, leaving the FCS to be decided by the two categorical ones; tolerances tied to the empirical distributions of ρ and m would widen their spread.

The study has limitations. The dataset is small and records approvals rather than repayment behaviour, so the classifier learns past lending practice, bias included. FBCE relies on proxies for liabilities and tenure, and its tolerances are heuristic. The benchmark uses default hyper-parameters and a single fold assignment, probabilities were not calibrated, and the dashboard has not yet been tested with loan officers.

VII. CONCLUSION AND FUTURE WORK

This paper presented FBCE, a loan decision-support prototype that shows a classifier's prediction, an independent financial consistency score and a SHAP explanation side by side. On the Loan Prediction dataset, logistic regression matched the ensembles (80.30% accuracy, 87.14% F1-score, AUC of 0.751), yet no model rejected even half of the historically rejected applications. In the case study, a borderline approval at 50.09% came with an FCS of 90.89, and SHAP revealed a gender effect that the label alone would have hidden.

Future work will use data with liabilities, employment tenure and repayment history, so that the FBCE tolerances and weights can be fitted rather than set by hand. We also plan to remove protected attributes, add fairness metrics and probability calibration, test a review-priority index that combines the FCS with the model probability, evaluate the dashboard with loan officers and expose the pipeline as a service.

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